iQuanta
Building Learning Networks

USEFUL FORMULAE & CONCEPT

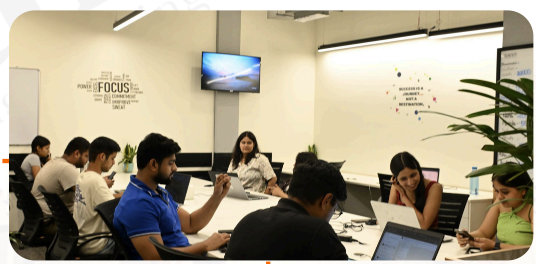
NUMBER SYSTEM

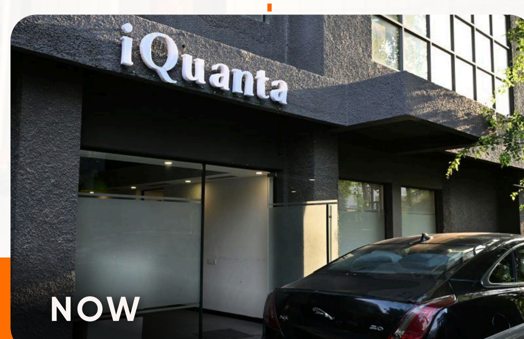
CAT QUANTITAIVE APTITUDE

About iQuanta?

Founded in 2015 by Indrajeet Singh, iQuanta began as a Facebook doubt-solving group from a small apartment. With strong expertise in Quants, he earned the title 'Wizard of Quant' by solving thousands of students' queries with clarity and speed. Noticing the lack of quality guidance for CAT aspirants, he officially launched iQuanta to bridge this gap.

Over the years, iQuanta has continued to support over 4 lakh students through its active online communities, offering resources, and peer-driven learning. In the last 5 years, more than 8,000 students have converted IIMs. In 2024, iQuanta produced 500+ 99 percentilers, many of whom made it to top IIMs. iQuanta continues to guide thousands of aspirants on their journey to top B-schools.





Indrajeet Singh
Founder & CEO, iQuanta
Forbes 30 Under 30

NUMBER SYSTEM

Important points to remember:

1. The product of 'n' consecutive natural numbers is always divisible by n!

2. Square of any natural number can be written in the form of 3n or 3n+1.

3. Square of a natural number can only end in 0, 1, 4, 5, 6 or 9. Second last digit of a square of a natural number is always even except when last digit is 6. If the last digit is 5, second last digit has to be 2.

4. Any prime number greater than 3 can be written as 6k+1 or 6k -1.

5. Any two-digit number 'mn' can effectively be written as 10m+n and a three-digit number 'lmn' can effectively be written as 100l+10m+n.

6. Sum of 1 st n natural number, $1+2+3+.....+n = \Sigma n = n(n+1)/2$

7. Sum of 1 st n positive even numbers, $2+4+6$ ------$+2n = n(n+1)$

8. Sum of 1 st n positive odd numbers, $1+3+5+$ ------$+(2n - 1) = n^2$

9. Sum of squares of 1 st n natural numbers :
$1^2+2^2+3^2+----+n^2 = n(n+1)(2n+1)/6$

10. Sum of cubes of 1 st n natural numbers :
$1^3+2^3+3^3+\text{--------}+n^3 = (\Sigma n)^2 = [n(n+1)/2]^2$

11. 0 and 1 are neither composite nor prime.

12. There are 25 prime numbers less than 100.

13. There are 15 prime numbers less than 50.

Divisibility rules

2 – when last digit of the number is 0, 2, 4, 6, 8.
4 – when last two digits of the number is 00 or divisible by 4.
8 - when the last three digits of the number is 000 or divisible by 8.
5 – when last digit is 0 or 5.
2^n - when last n digits of the numbers are divisible by 2^n.
5^n - when last n digits of the number are divisible by 5^n.

Divisibility rule for 3 and 9
3 – Sum of the digits should be divisible by 3.
9 – Sum of digits should be divisible by 9.

Divisibility rule for 7 and 11
7 - the difference between twice the unit digit of the given number and the remaining part of the given number should be a multiple of 7 or it should be equal to 0".
For example, 798 is divisible by 7.
11 - if the difference between the sums of the alternate digits of the given number is either 0 or divisible by 11, then the number is divisible by 11.

Divisibility rules of 6, 12, 14, 15, 18, etc.
Whenever we have to check the divisibility of a number N by a composite number C, the number N should be divisible by all the prime factors (the highest power of every prime factor) present in C.

Properties of factors and co-prime numbers

For any natural number N = $a^x * b^y * c^z$...., where a , b, c.... are prime numbers and x, y, z are their powers which are natural numbers. (N is prime factorized here).

Number of factors of N = (x+1) (y+1) (z+1) ....
Number of even factors = same as above, but just consider one power of 2 less while calculating.
Number of odd factors = (Total factors – even factors) OR remove 2 and its powers from the prime factorization & find factors of rest of the odd part.

Remainder and Remainder Theorems

Dividend = Divisor x Quotient + Remainder

Mod, which stands for modulus simply means 'remainder'. So, the value of a mod b is simply the remainder obtained on dividing a by b.

If Rem $[N_1/D]$ = R1, Rem $[N_2/D]$ = R2, ..., Rem $[N_m/D]$ = R_m then
i. Rem $[(N_1+N_2+....+N_m)/D]$ = Rem $[(R_1+R_2+....+R_m)/D]$
ii. Rem $[(N_1*N_2*....*N_m)/D]$ = Rem $[(R_1*R_2*....*R_m)/D]$

iii. **Negative Remainder:** Sometimes we use negative remainder to find the actual remainder easily. Let us try to understand it :-
We know that remainder when 19 is divided by 10 is 9 but we can also take the remainder as –1 which is (9 – 10).

Remainder Theorems:

1. Wilson's Remainder Theorem

For any prime number P, Rem $[\frac{(P-1)!}{P}]$, i.e.,
(P-1)! mod P = (P – 1) or – 1
E.g., Rem $[\frac{18!}{19}]$ = 18 or -1

2. Euler's Totient Theorem

For two coprime numbers N and D, Rem $[\frac{N^{k.E(D)}}{D}]$ = 1, where k is a natural number and E(D) is Euler of the number D. (Calculation of how to calculate Euler mentioned above).

Example: What is the remainder when 47^{32} is divided by 51?
Solution: 51 = 3∗17 so E (51) = 51* $\frac{2}{3}*\frac{16}{17}$ = 32.
We can see that 47 and 51 are coprime, so using Euler's remainder theorem,

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Finding the Last digit or Unit's Digit (UD) of a Number:

Unit's digit of a number, i.e., the last digit is the remainder when the number is divided by 10.

E.g. Unit's digit of 2345 is 2345 mod 10 = 5
Unit's digit of (234 + 567) = Unit digit of (4 + 7) = Unit Digit of 11 = 1
Similarly, to find the unit's digit of (234*567), we just need to consider the unit's digits of these two numbers viz: 4*7 = 28. Hence the unit's digit of the given product is 8.


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How to calculate the unit's digit of numbers of the form a^b such as 2^{253}, 3^{93}, 4^{74} etc.?

Case 1: When b is NOT a multiple of 4.
We find the remainder when b is divided by 4.
Let b = 4k + r, where r is the remainder when b is divided by 4, and
0 < r < 4.
The unit's digit of a^{4k+r} is the unit digit of a^r.

Case 2: When b is a multiple of 4.
We observe the following conditions:
Even numbers 2, 4, 6, 8 when raised to powers which are a multiple of 4 give the unit's digit as 6.
Odd numbers 3, 7, and 9 when raised to powers which are a multiple of 4 give the unit's digit as 1.
Unit's digit of a^{4k} = units digit of a^4

Finding the Last Two Digits (LTD) of a number:

Last two digits of a number is the remainder when the number is divided by 100.
E.g. LTD(123456) = 56, LTD(123*456) = LTD(23*56) = LTD(1288) = 88

How to calculate last two digits of numbers in the form a^b such as 24^{356}, 53^{903}, 79^{714} etc.?

Case 1: When the unit's digit of a is 1, multiply the ten's digit of the number with the last digit of the exponent to get the ten's digit and it is easy to understand that the unit's digit is equal to one.

Example: Find the last two digits of 21^{43}.
Solution: We know that unit's digit of the given number will be 1.
As mentioned above, to find the second last digit we need to multiply the ten's digit of the number (2) to the unit's digit of the exponent (3), 2*3 = 6.
So LTD of 21^{43} is 61.

Case 2: When a = 2^n while finding last two digits of a^b.
2^{10} = 1024, 24^1 = 24, 24^2 = 576, ..., 24^3 = xxx24, 24^4 = xxxxx76

Following the pattern, it can be noticed that,
LTD(24^{odd}) = 24 and LTD(24^{even}) = 76.

Example:
LTD(16^{125}) = LTD(2^{500}) = LTD(xx24^{50}) = LTD(24^{even}) = 76.

Case 3: When the unit's digit of a is 5 while finding the last two digits of a^b.
LTD(xxA5^B) = 75, when A and B are odd.
LTD(xxA5^B) = 25, otherwise.

LTD(35^{125}) = 75, LTD(45^{242}) = 25, LTD(85^{775}) = 25, LTD(75^{364}) = 25.

LCM, HCF AND THEIR APPLICATIONS

Important points to remember:

1. For two natural numbers x and y where x = h*a and y = h*b where a, b are co-prime numbers and HCF (x, y) = h, LCM (x, y) = h*a*b and Product (x, y) = h^2*a*b
It can be noticed that:
(i) Product (x, y) = LCM (x, y) * HCF (x, y)
(ii) LCM is always a multiple of HCF.

2. If N is a positive integer which leaves remainder 'r' each time when divided by x, y or z then N = {LCM (x, y, z) * k} + r, where k is a whole number.

E.g. Find the smallest number which leaves a remainder of 3 when divided by 4, 5 or 6.
Solution: LCM(4,5,6)*1 + 3 = 63

3. If N is a positive integer which leaves remainders r_1, r_2 and r_3 when divided by x, y and z respectively, where x - r_1 = y - r_2 = z - r_3 = r then

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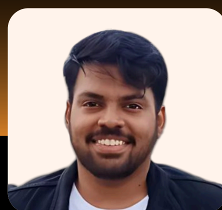

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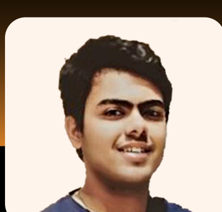

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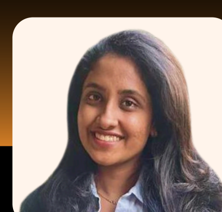

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And many more

N = {LCM (x, y, z) * k} - r, where k is a natural number.

E.g. Find the smallest number which when divided by 8, 9, 10 leaves remainders 3, 4, 5 respectively.
Solution: We see that 8-3 = 9-4 = 10-5 = 5
So, smallest number = LCM (8, 9, 10)*1 – 5 = 355

4. N is the greatest positive integer which divides x and y, leaving remainder r_1 and r_2 respectively, then N = HCF ($x-r_1$, $y-r_2$).

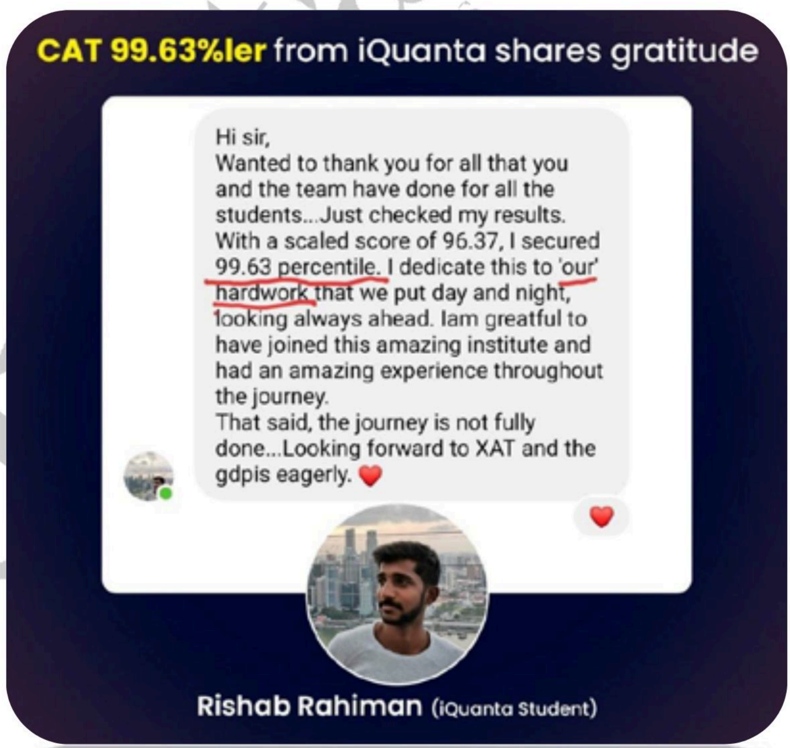
5. N is the greatest positive integer which divides x, y and z, leaving a remainder r in each case, then N = HCF (y-x, z-y).

Index of Greatest Power

The highest power of a number m that can divide another number n is known as IGP of m in n.

For any prime number p, IGP(p) in n! = $[\frac{n}{p}] + [\frac{n}{p^2}] + [\frac{n}{p^3}] + [\frac{n}{p^4}]$ + ------- where [x] denotes the greatest integer less than or equal to x. There is another way to get the IGP. It is shown in the example below.

Example: What is highest power of 2 that can divide 40!?
Solution: IGP of 2 in 40! = $[\frac{40}{2}] + [\frac{40}{2^2}] + [\frac{40}{2^3}] + [\frac{40}{2^4}] + [\frac{40}{2^5}]$ = 20+10+5+2+1 = 37
OR
$[\frac{40}{2}]$ --> $[\frac{20}{2}]$ --> $[\frac{10}{2}]$ --> $[\frac{5}{2}]$ --> $[\frac{2}{2}]$ --> 1
Answer = 20+10+5+2+1 = **37**



Why Students Love iQaunta

Yogyata Srivastava
it was a great session sir and really love your patience with us like answering every question
1h Like Reply

Simran Bahl
non eng modul helped a lot in todays class
1h Like Reply

Alefiya A
the session was really great! thank you.
1h Like Reply

Satyam Sharma
sir it was really mind blowing session...
also a quick question, do we need to rote memorize all the prime numbers?
because it will help us in Quick factorization and topics like divisibility, arithmetic
1h Like Reply

Anmol Verma
Great session.... Bhout saare concepts and doubts regarding Mixtures and Alligations solve hogye.....
👍👍👍👍
10h Like Reply

Bhavani Sri Sundaram
Alligations is that topic I didn't have confidence in. This is the third time revising this, got a hang of it now. I'm just mentally face palming myself. Thank you for the session sir.
10h Like Reply

Navonil Ghosh
Thank you sir. Bohot acha revision ho raha hai
10h Like Reply

Vanshika Gautam
Great Session sir !Thanku sir
1h Like Reply

Shivani Yadav
amazinggggggggggggggggggggggggggggggggg
1h Like Reply

Gowri Krishna
It was great and basic for us to understand! Thank you...please tell us what to do before the next class
1h Like Reply

Aditya Pandey
great session
1h Like Reply

Parvathi V Nair
hopefully i'll heal from my math trauma from school :')
1h Like Reply 1 😜

Anubhav Raj
Maaza aaya sir. Able to solve mostly all of the question. Thank you sir and good night 😴
10h Like Reply

Himanwita Mazumdar
Didn't realise the time today at all!

Jaya Anshu
If teachers in school taught like this , life would have been lot simpler
1h Like Reply

Shanmukha Rao
ALP sir class will never get bored
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Christopher Tarantino
Great session sir. Rather helpful and humorous. Thank you!
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Vishal Yadav
that a great session

Harsh Bhansali
Non engineering quants revised and some other mind boggling things
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Nyasa Mittal
great session sir

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